momentum eigenstate is uniformly spread over the entire *x* axis. To see this, let's take the wave function in Eq. 8.17 and multiply it by its complex conjugate:

$$\psi_p^*(x)\psi_p(x) = \left(\frac{1}{\sqrt{2\pi}}e^{\frac{-ipx}{\hbar}}\right)\left(\frac{1}{\sqrt{2\pi}}e^{\frac{ipx}{\hbar}}\right) = \frac{1}{2\pi}.$$

The result is completely uniform, with no peaks anywhere on the *x* axis. Evidently, a state with definite momentum is completely uncertain in its position.

Fig. 8.2 illustrates the definition of uncertainty for the position variable *x*. In the top half of the figure, you can see that the uncertainty Δx is a measure of how spread out the function is in relation to its expectation value $\langle x \rangle$. The label *d* shows the deviation of one point in relation to $\langle x \rangle$; this may be a positive or negative quantity. The uncertainty Δx is the result of an averaging process over all possible *d*'s and characterizes the function as a whole. To prevent the positive *d*'s from canceling the negative ones, each *d* value is squared during this averaging process.

The bottom half of Fig. 8.2 shows how the calculation can be simplified by shifting the origin to coincide with $\langle x \rangle$. The numerical value of Δx is unchanged by this shift.